

Loop Idioms: What We Do in Loops

Note: Even though these examples are simple,
the patterns apply to all kinds of loops

Making “smart” Loops

The trick is “knowing” something about the whole loop when you are stuck writing code that only sees one entry at a time

Set some variables to initial values

for **thing** **in** **data**:

Look for something or do something to each entry separately, updating a variable

Look at the variables

Looping Through a Set

```
print('Before')
for thing in [9, 41, 12, 3, 74, 15] :
    print(thing)
print('After')
```

```
$ python basicloop.py
```

Before

9

41

12

3

74

15

After

What is the Largest Number?

What is the Largest Number?

3

What is the Largest Number?

41

What is the Largest Number?

12

What is the Largest Number?

9

What is the Largest Number?

74

What is the Largest Number?

15

What is the Largest Number?

What is the Largest Number?

3

41

12

9

74

15

What is the Largest Number?

largest_so_far

-1

What is the Largest Number?

3

largest_so_far

3

What is the Largest Number?

41

largest_so_far

41

What is the Largest Number?

12

largest_so_far

41

What is the Largest Number?

9

largest_so_far

41

What is the Largest Number?

74

largest_so_far

74

What is the Largest Number?

15

74

What is the Largest Number?

3 41 12 9 74 15

74

Finding the Largest Value

```
largest_so_far = -1
print('Before', largest_so_far)
for the_num in [9, 41, 12, 3, 74, 15] :
    if the_num > largest_so_far :
        largest_so_far = the_num
    print(largest_so_far, the_num)

print('After', largest_so_far)
```

```
$ python largest.py
```

```
Before -1
```

```
9 9
```

```
41 41
```

```
41 12
```

```
41 3
```

```
74 74
```

```
74 15
```

```
After 74
```

We make a variable that contains the largest value we have seen so far. If the current number we are looking at is larger, it is the new largest value we have seen so far.

More Loop Idioms



Acknowledgements / Contributions



These slides are Copyright 2010- Charles R. Severance
(www.dr-chuck.com) of the University of Michigan School of
Information and open.umich.edu and made available under a
Creative Commons Attribution 4.0 License. Please maintain this
last slide in all copies of the document to comply with the
attribution requirements of the license. If you make a change,
feel free to add your name and organization to the list of
contributors on this page as you republish the materials.

Initial Development: Charles Severance, University of Michigan
School of Information

... Insert new Contributors and Translators here

...